

CHAPTER COMPACT REVISION NOTES: INTRODUCTION TO POLYNOMIALS

1. Core Concepts & Terminology

- **Polynomial:** An algebraic expression consisting of variables, constants, and exponents, combined using mathematical operations (+, -, ×). The powers of the variables must strictly be **non-negative integers** (i.e., 0, 1, 2, 3, ...).
Example: $3x^2 - 4x + 5$ is a polynomial.
Non-Example: $2x^{-1}$ or $\sqrt{x}$ are **not** polynomials because their exponents are not non-negative integers.
- **Terms:** The parts of the polynomial expression separated by + or - signs.
- **Coefficient:** The numerical factor multiplied by a variable in a term.
Example: In $3x^2$, the coefficient of x^2 is **3**.
- **Degree:** The **highest exponent (power)** of the variable in the polynomial expression.
Example: The degree of $3x^7 - 4x^6 + x + 9$ is **7**.

2. Classification of Polynomials

By Number of Terms:

- **Monomial:** A polynomial containing exactly **one** term (e.g., $5x$, 3).
- **Binomial:** A polynomial containing exactly **two** terms (e.g., $x + 2$, $3x^2 - 5$).
- **Trinomial:** A polynomial containing exactly **three** terms (e.g., $x^2 + 3x + 2$).

By Degree:

Polynomial Type	Degree	Standard Form	Example
Constant Polynomial	0	$p(x) = c$	5, -3
Linear Polynomial	1	$p(x) = ax + b$	$2x + 1$
Quadratic Polynomial	2	$p(x) = ax^2 + bx + c$	$x^2 - 4x + 7$
Cubic Polynomial	3	$p(x) = ax^3 + bx^2 + cx + d$	$3x^3 + 2x^2 - x + 5$

(Note: A **Zero Polynomial** is simply $p(x) = 0$. Its degree is **undefined**.)

3. Value and Zeroes of a Polynomial

- **Value of a Polynomial:** Obtained by replacing the variable with a specific real number. The value of $p(x)$ at $x = k$ is denoted as $p(k)$.
- **Zero (or Root) of a Polynomial:** A real number α is called a zero of a polynomial $p(x)$ if and only if $p(\alpha) = 0$.
- **Geometrical Meaning:** The zeroes of a polynomial $p(x)$ correspond to the **x-coordinates of the points where its graph intersects the x-axis**.
- A *Linear polynomial* graph is a straight line and cuts the x-axis at exactly **1 point**.
- A *Quadratic polynomial* graph is a parabola ($\cup$ or $\cap$) and can cut the x-axis at a maximum of **2 points**.
- **General Rule:** A polynomial of degree n can have at most **n real zeroes**.

4. Key Algebraic Theorems

Remainder Theorem: If a polynomial $p(x)$ of degree ≥ 1 is divided by a linear polynomial $(x - a)$, then the **Remainder is equal to $p(a)$** .

Factor Theorem: For a polynomial $p(x)$:

1. If $p(a) = 0$, then $(x - a)$ is a **factor** of $p(x)$.
2. Conversely, if $(x - a)$ is a factor of $p(x)$, then **$p(a) = 0$** .

5. Important Algebraic Identities

- $(x + y)^2 = x^2 + 2xy + y^2$
- $(x - y)^2 = x^2 - 2xy + y^2$
- $x^2 - y^2 = (x - y)(x + y)$
- $(x + a)(x + b) = x^2 + (a + b)x + ab$
- $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$
- $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$
- $(x - y)^3 = x^3 - y^3 - 3xy(x - y)$
- $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$
- *Special Case:* If $x + y + z = 0$, then **$x^3 + y^3 + z^3 = 3xyz$** .